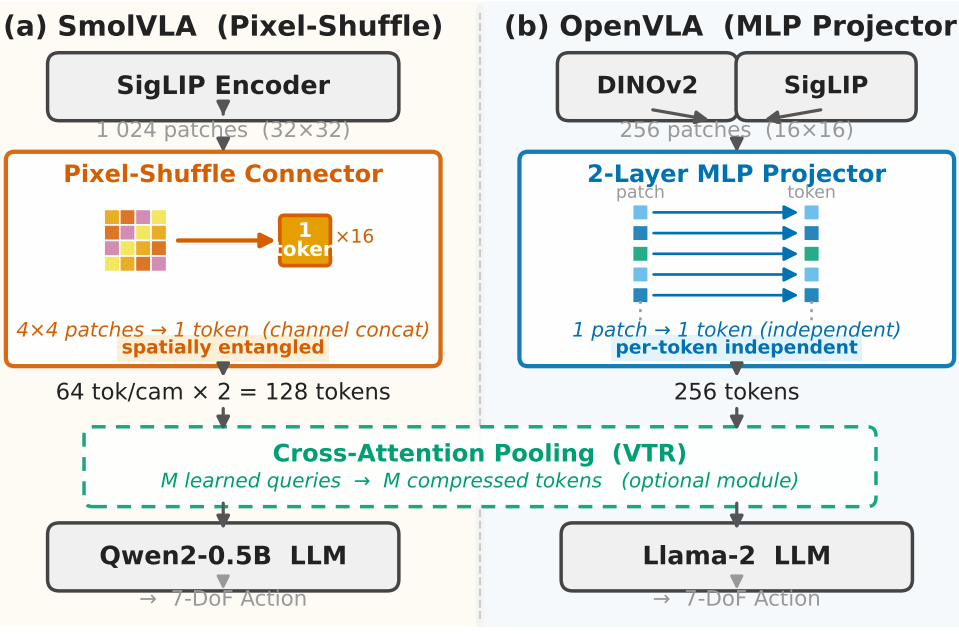
(a) SmolVLA (Pixel-Shuffle)
SigLIP Encoder
1 024 patches (32×32)
Pixel-Shuffle Connector
1 token
×16
4×4 patches → 1 token (channel concat)
spatially entangled
64 tok/cam × 2 = 128 tokens
(b) OpenVLA (MLP Projector)
DINOv2
SigLIP
256 patches (16×16)
2-Layer MLP Projector
patch
token
1 patch → 1 token (independent)
per-token independent
256 tokens
Cross-Attention Pooling (VTR)
M learned queries → M compressed tokens (optional module)
Qwen2-0.5B LLM
→ 7-DoF Action
Llama-2 LLM
→ 7-DoF Action